
IMPROVING IMAGE CLASSIFICATION WITH RESNET18

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Department of Computer Science
Aarhus University

Department of Computer Science
Aarhus University

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ABSTRACT

This project investigates the performance of a ResNet-18 model on a multi-class sports image classification task and evaluates how targeted training optimizations affect accuracy and generalization. A baseline model using standard transfer learning is compared with an optimized model incorporating data augmentation, regularization (weight decay and Mixup), learning-rate scheduling, and class balancing through weighted sampling. Experiments are conducted across three training-set sizes (100%, 10%, and 5%) to assess robustness under varying data constraints. Results show that the optimized model consistently outperforms the baseline, yielding higher validation accuracy, more stable loss trajectories, and substantially improved performance on the weakest classes—most notably in the highly challenging shooting category. The performance gap widens as the dataset shrinks, demonstrating that the introduced techniques are particularly valuable in low-data settings. Overall, the study highlights how carefully selected training strategies can substantially improve model reliability and class-wise fairness in real-world, imbalanced image classification tasks.

Keywords Sports Image Classification · Optimization · ResNet

1 Introduction

Deep learning has become a central approach for solving image recognition tasks, with convolutional neural networks (CNNs) demonstrating state-of-the-art performance across a wide range of domains such as natural images, medical imagery, and autonomous systems. Among modern CNN architectures, residual networks (ResNets) have proven particularly effective due to their skip-connection design, which enables stable training of deeper models and improved generalization. Transfer learning has further strengthened these models by allowing pre-trained feature extractors to be adapted efficiently to new tasks with limited data. In this project, we investigate the use of ResNet18 for multi-class image classification on a sports-image dataset containing 22 different sports categories. The dataset includes substantial visual diversity—variation in lighting, viewpoints, backgrounds, and motion—as well as naturally occurring class imbalance. These factors make the classification task challenging and provide a realistic setting for evaluating model robustness.

Although ResNet18 is widely used in practice, its performance depends heavily on the choice made during training: optimization settings, data preprocessing, regularization, and handling of dataset imbalance. Poor hyperparameter choices or insufficient regularization can lead to issues such as overfitting, noisy learned features, or model bias toward majority classes. Because of this, simply applying transfer learning is not guaranteed to yield optimal performance. The goal of this project is to evaluate the performance of a ResNet18 model on the sports-image dataset and optimize the training process to improve classification accuracy and robustness. To achieve this, we systematically apply several regularization and balancing techniques to optimize the model which we then compare to a baseline ResNet18 model using only transfer learning. These models are compared in three different training regimes — using 100%, 10%, and 5% of the available training data — to assess how well each approach generalizes under varying levels of data scarcity.

2 Related Work

2.1 Data Augmentation

The efficiency of large deep convolutional neural networks (CNN's) is often dependent on their ability to manage the vast number of parameters inherent in their architecture, which can lead to significant overfitting. The easiest and most common method used on image data to address this problem is data augmentation, according to Krizhevsky et al. [4]. "Through a combination of spatial and geometric augmentations, the pipeline first resized all images to 256 X 256 and then extracted random 224 X 224 crops to introduce variation in scale and composition. Additional photometric augmentations—such as color jitter and other RGB intensity perturbations—were applied to simulate natural variations in lighting and contrast. Together, these transformations effectively expanded the diversity of the training data, exposing the model to a broader range of visual patterns and thereby reducing overfitting while improving generalization performance."

2.2 Batch Normalization

In 2015 Lofe and Szegedy [2] proposed a way to reduce the *internal covariate shift* using batch normalization (BA). BA is a widely used technique that stabilizes and enables faster training of deep neural networks. It happens through normalizing the activations across mini-batches. Challenges with saturating activation functions, such as sigmoid and tanh, were the original motivation behind the method. Furthermore, on ReLU architectures the method has been highly effective since BA stabilizes training. By stabilizing activation statistics, BA improves gradient flow and reduces sensitivity to initialization, enabling the use of significantly higher learning rates without destabilizing training. Additionally, mini-batches stochastic nature provides a mild regularizing effect, often reducing the need for techniques such as dropout.

In general, BA represents a standard component in modern image classification models with its important advancement in deep model training efficiently and robustly. With its stabilizing effects, compatibility with ReLU networks and its ability to support higher learning rates, it is a central technique in contemporary image classification.

2.3 ResNet & ADAM)

BA was used by He et al. [1] where it comes after every convolutional layer, maintaining stable activation distributions and supports efficient optimization even in very deep models. The main aspect of the paper is addressing a critical limitation in deep CNNs: the *degradation problem*, where the network depth increases and the accuracy gets saturated. To overcome this, ResNets reformulate the learning task as *residual learning*. The core insight of the *deep residual learning* framework is to explicitly let each of the stacked layers fit a residual mapping. The layers are optimized to approximate a residual function $F(x) := H(x) - x$ instead of the desired mapping $H(x)$. They also use identity shortcut connections to add inputs directly to outputs, as seen in Figure 1

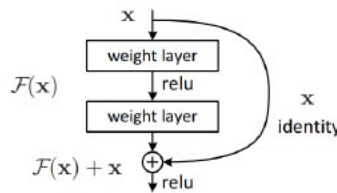


Figure 1: Residual learning: a building block

To do their findings, they used an optimizer called momentum-based SGD to regularize the architecture and stabilize optimization. In this project, we will be using an ADAM optimizer [3] as the default choice because it often works well with a constant learning rate due to its update rules, which produce step sizes that are approximately bounded by the base learning rate. In addition to the constant learning rate, ADAM can also be used with adaptive learning rates which allows large flexibility in a project like this one .

3 Methods

3.1 Data

The dataset consists of images representing 22 different sports categories, such as basketball, shooting, weightlifting, and swimming.

Each image belongs to exactly one sport class, making the task a standard single-label multi-class classification problem. Images vary in content, background, camera angle, and lighting conditions, and the number of samples per class is unevenly distributed. This imbalance makes the dataset more challenging, as some sports (e.g., cricket or baseball) appear more frequently than others (e.g., shooting or kabaddi). All images were stored as RGB files in varying resolutions. Before training, the dataset was split into a 60% training set and 40% validation set, using stratified sampling to preserve the original class distribution. A few samples contained corrupted or unreadable files - these were filtered out automatically during loading.

3.2 Preprocessing

Since we fine-tuned a pretrained ResNet-18, it was crucial to ensure that inputs matched the network’s expected format. All images were resized to 224 X 224 pixels, following the input specification of ImageNet-pretrained models. To ensure compatibility with the pretrained weights, pixel intensities were normalized using the ImageNet channel-wise means and standard deviations

3.3 Model Architecture and Transfer Learning

For all experiments, we used a ResNet18 architecture pre-trained on ImageNet. Transfer learning was chosen because the dataset consists of natural RGB photographs, making ImageNet features a suitable starting point. Using a pre-trained model allowed us to train effectively with limited data while avoiding the need to learn low-level visual features (edges, corners, color blobs) from scratch. We replaced the final fully connected layer of ResNet18 with a new classifier matching the number of sport classes in the dataset while all other layers were initialized with ImageNet weights.

3.4 Overfitting and Data Augmentation

Initial experiments using the basic training pipeline (no augmentation, no regularization) showed a clear pattern of overfitting: the training loss decreased steadily while the validation loss flattened or increased. This indicated that the model was memorizing training examples rather than learning generalizable representations.

To counter this, we introduced data augmentation to increase the effective size and diversity of the training set. Data augmentation artificially expands the training distribution by applying random transformations to each image, ensuring that the model encounters new variations of the same underlying images at every epoch. In practice, this makes the classification task harder in a controlled way, encouraging the model to learn generalizable features rather than memorize images during training. The following random transformations were used: Horizontal flipping, Small random rotations, Color jitter (brightness, contrast), Random cropping and Random erasing.

3.5 Regularization

To further reduce overfitting beyond data augmentation, the optimized model employed two additional regularization techniques: **weight decay** and **mixup**. Weight decay (L2 regularization) penalizes large parameter values during optimization, preventing the network from forming overly complex solutions that fit noise in the training data. This encourages smoother parameter updates and improves generalization. Mixup provides complementary data-level regularization by blending pairs of training images and their labels which results in smoother decision boundaries and improves robustness, especially in low-data settings.

3.6 Class Imbalance and Weighted Sampling

Exploratory analysis of the label distribution revealed substantial class imbalance, with some sports being represented by many more images than others. Without correction, the model tended to favor the majority classes, reflected both in validation accuracy and in confusion matrices.

To mitigate this, we applied a WeightedRandomSampler to the training set where each sample was assigned a weight inversely proportional to the frequency of its class. This oversampled minority classes and undersampled majority classes, leading to a more balanced sequence of training batches.

3.7 Training and evaluation

All models were trained using mini-batch gradient descent with the Adam optimizer and cross-entropy loss. Training proceeded for a fixed number of epochs, and after each epoch, the model was evaluated on the validation set using the same preprocessing pipeline (but without random augmentations). Validation loss and accuracy were then recorded to monitor generalization.

To ensure fair comparison between models, the same train-validation split and batch size were used throughout. During training, the model parameters were updated only on the training set, while validation data remained untouched. After each epoch, the validation accuracy was compared to the best score obtained so far. Whenever an improvement occurred, the model’s weights were saved. At the end of training, the model was restored to these best-performing weights. This strategy prevents late-epoch overfitting from negatively affecting the final reported performance and was used as a practical alternative to early stopping.

The baseline model was trained with a fixed learning rate and no regularization. In contrast, the optimized model incorporated the additional components described above. These were evaluated under identical conditions to isolate their impact on performance. Model quality was assessed using validation accuracy as the primary metric, complemented by training and validation loss in addition to per-class accuracy to better understand differences between well-represented and minority classes.

4 Results

Figure 2 shows the training and validation loss curves for the baseline and optimized models when trained on the full dataset. Both models converge stably, but their behaviors differ substantially in terms of generalization.

The baseline model exhibits rapid reduction in training loss, approaching near-zero values after only a few epochs. However, its validation loss plateaus around 0.39 and begins to fluctuate in later epochs, with a sharp increase after epoch 15. This pattern indicates overfitting: the model continues to memorize training examples even after generalization performance has stopped improving. The best validation accuracy achieved by the baseline model was **0.8968**.

In contrast, the optimized model shows smoother and more consistent convergence. Training loss decreases gradually throughout all epochs, and validation loss continues to improve well into the second half of training, eventually reaching a minimum of **0.3342**. Unlike the baseline model, the optimized model does not exhibit late-epoch degradation, demonstrating improved stability and resistance to overfitting. The best validation accuracy achieved was **0.9095**, representing a clear improvement over the baseline.

4.1 Per-class comparison

An examination of per-class accuracy, seen in Table 1 revealed substantial differences in how the two models generalized across the 22 sports categories. The baseline model performed well overall but showed clear weaknesses on several classes, with its lowest accuracies falling between 76% and 86%. In contrast, the optimized model achieved a much more consistent performance profile: its lowest per-class accuracies ranged from 82% to 87%, and no class fell below the 80% mark. This narrowing of the performance gap—particularly for the previously weakest classes—indicates that the optimized training pipeline not only improved overall accuracy but also substantially enhanced robustness across categories.

Model	Lowest Class	Accuracy	Next Two Lowest (Accuracy)
Baseline	Hockey	0.760	Tennis (0.852), Badminton (0.865)
Optimized	Hockey	0.822	Badminton (0.857), Gymnastics (0.873)

Table 1: Lowest per-class accuracies for the baseline and optimized models (100% training data).

4.2 Reducing size of dataset

To further evaluate the robustness of each training pipeline, we also conducted experiments using substantially reduced training sets (10% and 5% of the original training set). Deep learning models often perform well when abundant data and fully supervised signals are available, but their generalization ability can degrade sharply when data becomes limited. The following sections therefore compare baseline and optimized models when trained on 10% and 5% of the dataset, highlighting how each approach adapts to increasingly challenging learning scenarios.

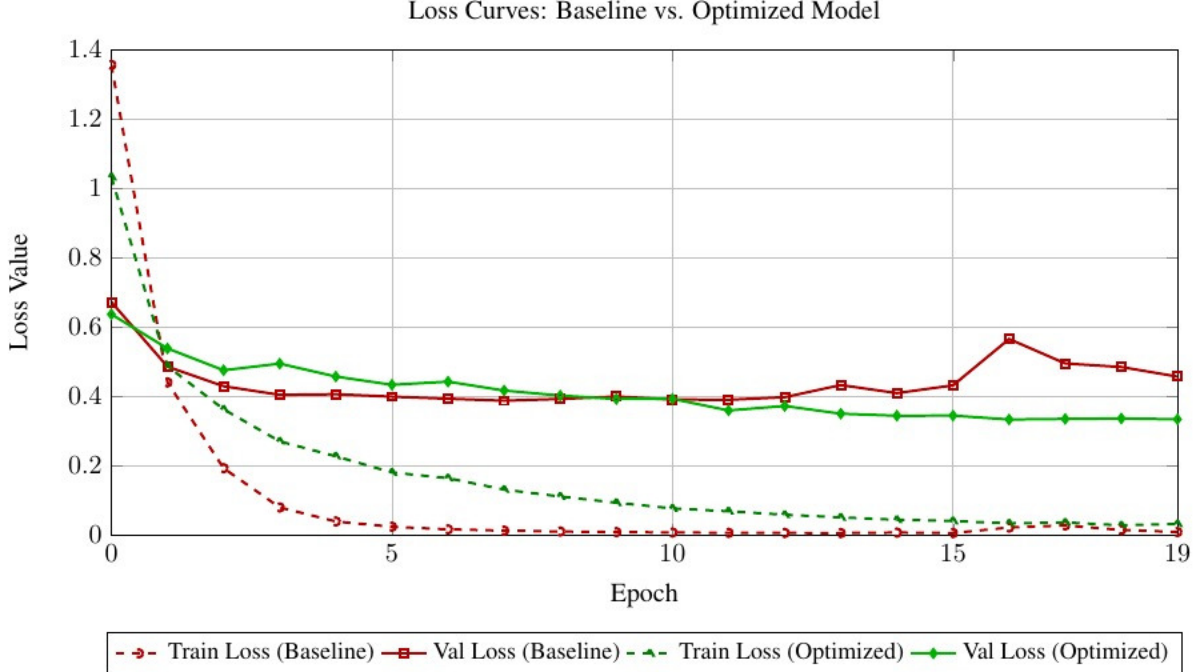


Figure 2: Loss Comparison

4.2.1 10% dataset

Reducing the training set to 10% led to a substantial drop in performance for both models, as expected in a low-data regime. The baseline model reached a peak validation accuracy of **74.7%**, with validation loss decreasing steadily across training but stabilizing at a relatively high level. The optimized model improved upon this result, achieving a higher peak validation accuracy of **77.1%**, along with consistently lower validation loss throughout most of training.

The learning curves reveal clearer distinctions between the models under limited data. The baseline model exhibits a strong train-validation gap from the early epochs, indicating overfitting despite the small dataset. In contrast, the optimized model shows a more gradual reduction in training loss and a tighter alignment with validation loss. This suggests that the added regularization—data augmentation, Mixup, weight decay and scheduled learning rate—plays a more influential role when data is scarce.

The per-class accuracy results (Table 2) further highlight the robustness of the optimized model. With the baseline model, the poorest-performing classes—most notably shooting (36.5%), Class gymnastics (57.4%), and Class tennis (59.0%)—show severe degradation, reflecting the model’s difficulty in generalizing from few examples. The optimized model reduces the severity of these failures: while some classes remain challenging (e.g., tennis at 57.2% and hockey at 62.2%), the lowest accuracies are consistently higher than those of the baseline model, and the majority of classes show meaningful improvements. Particularly notable is the improvement in classes that were weak in the baseline setting, such as shooting (from 64.4% to 65.4%) and badminton (from 74.9% to 82.2%).

Model	Lowest Class	Accuracy	Next Two Lowest (Accuracy)
Baseline	Shooting	0.365	Gymnastics (0.574), Tennis (0.590)
Optimized	Tennis	0.572	Hockey (0.622), Shooting (0.654)

Table 2: Lowest per-class accuracies for the baseline and optimized models (10% training data).

Overall, the 10% training-set experiments demonstrate that the optimized model not only achieves higher overall accuracy but is also substantially more stable across classes. The improvements are more pronounced than in the full-data setting, confirming that regularization techniques provide greater benefit when training data is limited.

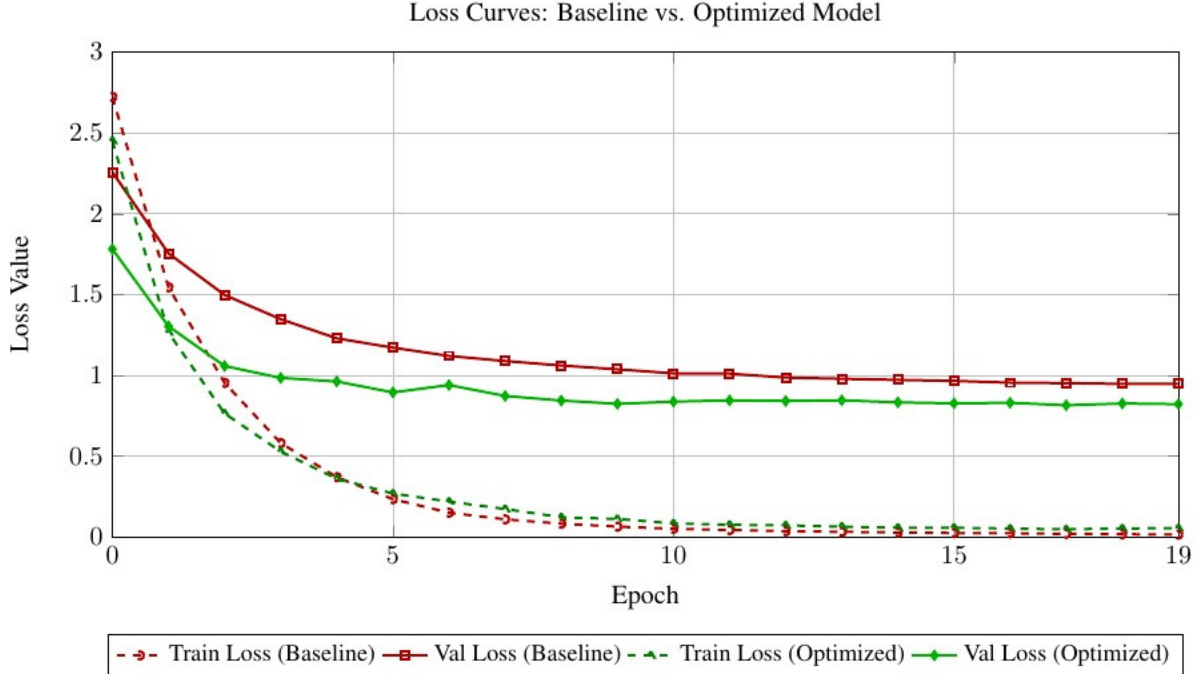


Figure 3: Loss Comparison

4.2.2 5% dataset

The baseline model struggled significantly under this low-data regime. Its validation accuracy plateaued at 67.3%, and—even though the training loss decreased rapidly—the validation loss remained high throughout training. This widening gap between the two curves indicates severe overfitting: the model was able to memorize the small training set but failed to generalize.

The optimized model delivered a clear improvement, even though training conditions were very limited. Its validation accuracy increased to 70.96%, a gain of nearly +4 percentage points over the baseline. Both the training and validation curves show a smoother, more stable descent, and the validation loss decreases substantially more than in the baseline model, indicating improved generalization.

Per-class accuracy improves across nearly all categories. Importantly, the previously failing shooting class jumps from 3.8% to 11.4%, and most other visually ambiguous classes also show meaningful gains. While some categories remain challenging (e.g., *gymnastics*, *hockey*), the overall distribution is more balanced, suggesting that the optimized approach helps stabilize performance under extreme data scarcity.

Model	Lowest Class	Accuracy	Next Two Lowest (Accuracy)
Baseline	Shooting	0.038	Basketball (0.464), Hockey (0.493)
Optimized	Shooting	0.114	Gymnastics (0.528), Hockey (0.542)

Table 3: Lowest per-class accuracies for the baseline and optimized models (5% training data).

5 Discussion

This project set out to evaluate how a standard transfer-learning baseline compares to an optimized ResNet18 training pipeline across different data-availability regimes. Across all experiments—100%, 10%, and 5% training data—the results consistently show that the optimized model provides better generalization, improved stability during training, and more balanced per-class performance.

A central observation is that the full dataset appears relatively easy for a pretrained ResNet18, as indicated by the baseline model already achieving almost 90% accuracy with minimal regularization (Figure 2).

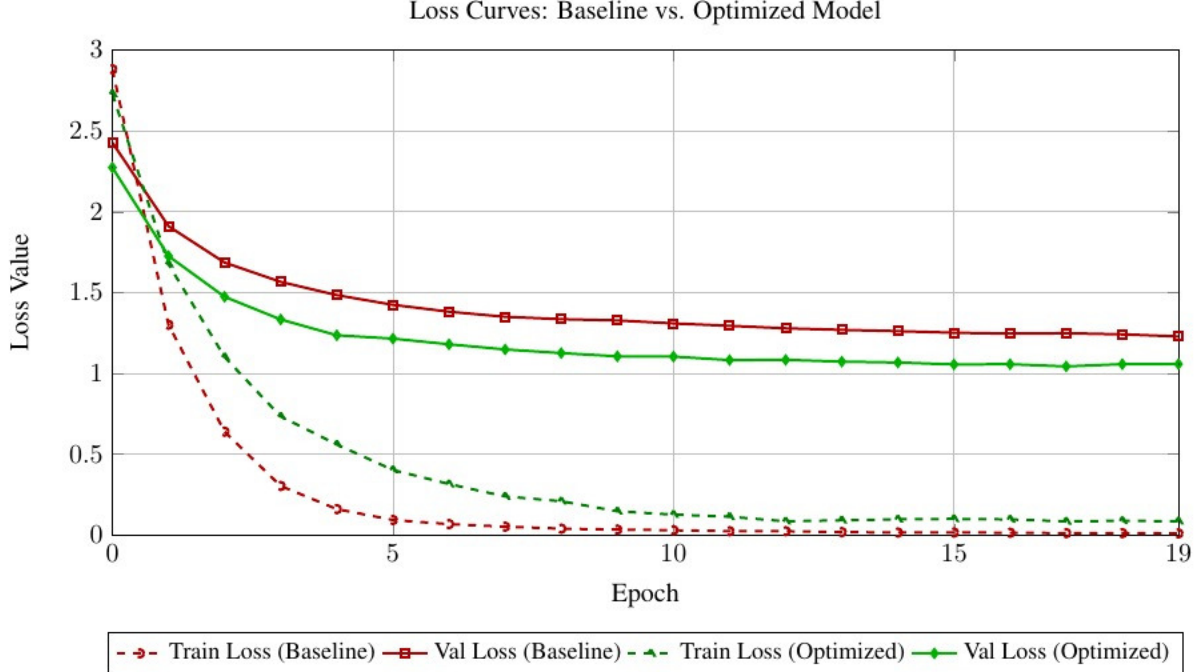


Figure 4: Loss Comparison

This ceiling effect likely limits the observable absolute gains of more advanced techniques. Nonetheless, the optimized model still provided clear improvements: lower validation loss, reduced overfitting, and higher minimum per-class accuracy. These effects became even more pronounced as the training set shrank.

The results from the 10% and 5% data regimes (Figures 3–4) highlight the role of regularization most strongly. In these low-data settings, the baseline model suffered from significant overfitting, reflected both in widening train–validation gaps and sharp drops in accuracy for several minority or visually complex classes. In contrast, the optimized model—benefiting from augmentation, mixup, weight decay, and weighted sampling—remained considerably more robust. Not only did it achieve higher overall accuracy, but it also mitigated extreme failure cases such as the *shooting* class, which improved from **3.8%** to **11.4%** in the 5% experiment. These findings suggest that regularization benefits scale inversely with dataset size, aligning with theoretical expectations in deep learning.

Despite the improvements, several limitations became apparent. First, the dataset’s relative simplicity for a modern CNN likely prevents a fuller assessment of the optimization techniques. A more challenging dataset—either with higher intra-class variation or more subtle visual distinctions—may reveal larger performance gaps. Second, the sports categories are unequally represented, and while weighted sampling helped, it does not fully address underlying distributional biases. Third, the study focused on a single architecture, and thus it remains unclear whether the same conclusions generalize to deeper models (e.g., ResNet50) or lighter ones suitable for deployment.

There are also methodological improvements worth considering. The experiments used a fixed training schedule without early stopping, which may have left unrealized performance gains, particularly for the baseline, which showed late-epoch degradation. Similarly, exploring more advanced learning-rate schedulers or label-smoothing could further increase robustness, especially in low-data settings. Finally, evaluating model confidence, misclassification patterns, or calibration would offer a more complete understanding of model behavior beyond accuracy alone.

Overall, this work demonstrates that a systematic training pipeline combining data augmentation, mixup, weight decay, and class-balanced sampling meaningfully improves robustness, especially when data is scarce. Even though the dataset limits the observable headroom for improvement, the optimized model consistently outperforms the baseline across all settings. Future studies could investigate these techniques on more challenging or diverse visual domains, or to pair them with modern training strategies such as self-supervised learning, to build an even deeper understanding of how to train reliable sports-image classifiers.

References

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